

Question1: Answer with TRUE or FALSE, don't justify your answers?

(7 M)

- 1) Significant digits are only important when rounding final results, not during intermediate calculations. **F**
- 2) Horner's algorithm is used for efficiently evaluating polynomials. **T**
- 3) The Taylor series converges rapidly for points far from the expansion point. **F**
- 4) The Bisection method guarantees finding the exact root in a finite number of steps. **F**
- 5) Rounding replaces a number with an approximation that minimizes error, whereas chopping discards digits without adjustment. **True**
- 6) The Mean-Value Theorem is a special case of Taylor's Theorem with zero derivatives. **F**
- 7) The error in the Bisection method after n steps is at most (b - a) divided by 2^n . **F**

Question 2: Given the following function:

(3 + 5 M)

$$f(x) = 22 - 4x^2 + 3x^3$$

- a) Use Horner's algorithm to evaluate $f(x)$ at $x = -2$
- b) What is the Taylor series for the function $f(x)$ about $c = -3$?

a) **3**

$$\begin{array}{r} 3 \quad -4 \quad 0 \quad 22 \\ -2 \quad \quad -6 \quad 20 \quad -40 \\ \hline 3 \quad -10 \quad 20 \quad -18 \end{array}$$

2

Thus, we obtain $P(-2) = -18$.

1

b) **5**

$$\begin{array}{r} 3 \quad -4 \quad 0 \quad 22 \\ -3 \quad \quad -9 \quad 39 \quad -117 \\ \hline 3 \quad -13 \quad 39 \quad -95 \\ \quad \quad -9 \quad 66 \\ \hline 3 \quad -22 \quad 105 \\ \quad \quad -9 \\ \hline 3 \quad -31 \end{array}$$

1

1

1

Thus, we can write:

$$P(x) = 3(x + 3)^3 - 31(x + 3)^2 + 105(x + 3) - 95$$

2

! Student can answer (b) by the successive derivatives of the function.

Question 3: Floating-point representation**(7 M)**

Determine the single-precision machine representation in a 32-bit word-length computer of the following decimal number (-23.21875) .

Solution:

1. Converting the integer part to binary, we have:

$$(23)_{10} = (27)_8 = (10111)_2.$$

1

2. Converting the fractional part to binary, we have:

$$(.21875)_{10} = (.16)_8 = (.001110)_2.$$

1

3. $(23.21875)_{10} = (10111.001110)_2 = (1.0111001110)_2 \times 2^4$ is the corresponding one-plus form in base 2, and $(.0111001110)_2$ is the stored mantissa.

1

4. Next the exponent is $(4)_{10}$, then $c - 127 = 4$, we immediately see that $(131)_{10} = (203)_8 = (1000\ 0011)_2$ is the stored exponent.

2

Thus, the single-precision machine representation of (-23.21875) is:

$$[1\ 10000011\ 0111001110\dots0000]_2$$

1

$$[1100\ 0001\ 1011\ 1001\ 1100\ \dots\ 0000]_2 = [C1B9C000]_{16}$$

1

Question 3: Given the following function:**(2 + 6 M)**

$$f(x) = -x^2 + 4x - e^x$$

- a) Prove that the function $f(x)$ has exactly one real root α in the interval $[0,1]$. 2
- b) Find a root of the function $f(x)$ on the interval $[0, 1]$ by using the bisection method. Terminate the iterations if they exceed $n_{max} = 2$ or if the error bound is less than 0.5×10^{-6} .

Solution:

a.

$$u = f(0) = -1$$

$$v = f(1) = 0.2817$$

Since, $u \cdot v < 0$ then the function f has exactly one real root α in the interval $[0, 1]$. 2

b.

1) $n = 0$ and $x_0 = (0 + 1)/2 = 0.5$, $w = f(x_0) = f(0.5) = 0.1013$

$u \cdot w < 0$ then: $a = a$, $b = x_0$ and the new interval is $[0, 0.5]$. Error = 0.5. 2

2) $n = 1$ and $x_1 = (0 + 0.5)/2 = 0.25$, $w = f(x_1) = f(0.25) = -0.3465$

$u \cdot w > 0$ then: $a = x_1$, $b = b$ and the interval is $[0.25, 0.5]$. Error = 0.25. 2

3) $n = 2$ and $x_2 = (0.25 + 0.5)/2 = 0.375$, $w = f(x_2) = f(0.375) = -0.0956$

$u \cdot w > 0$ then: $a = c$, $b = b$ and the interval $[0.375, 0.5]$. Error = 0.25. 2

n	x_n	$f(x_n)$	Error
0	0.5	0.1013	0.5
1	0.25	-0.3465	0.25
2	0.375	-0.0956	0.125